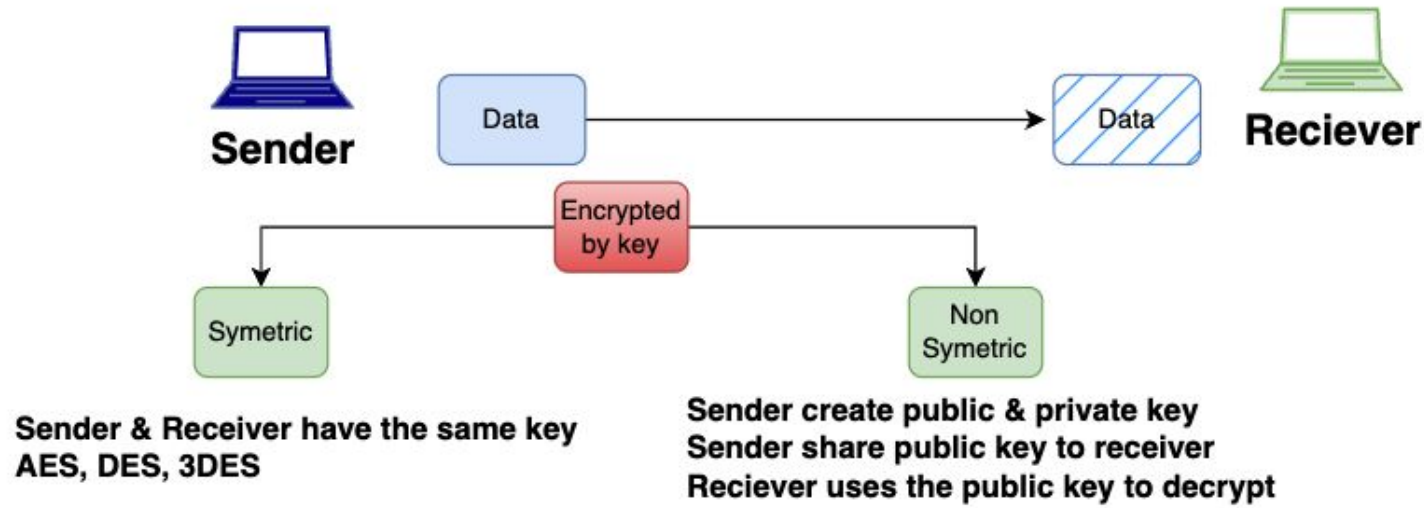


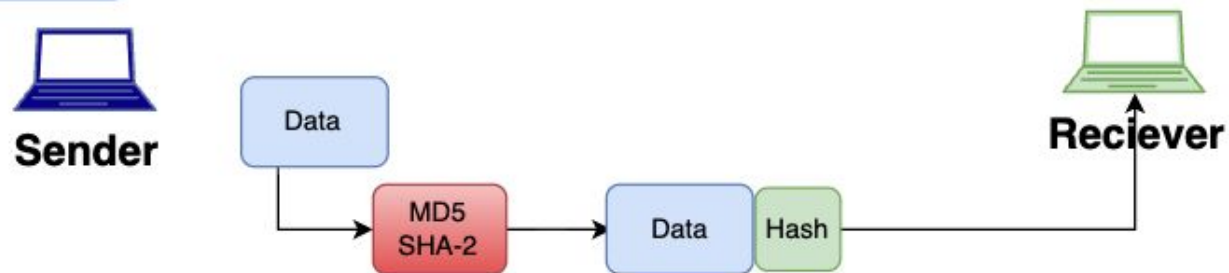
Confidentiality

- Make sure that the traffic is encrypted & opened only by authorised device



Integrity

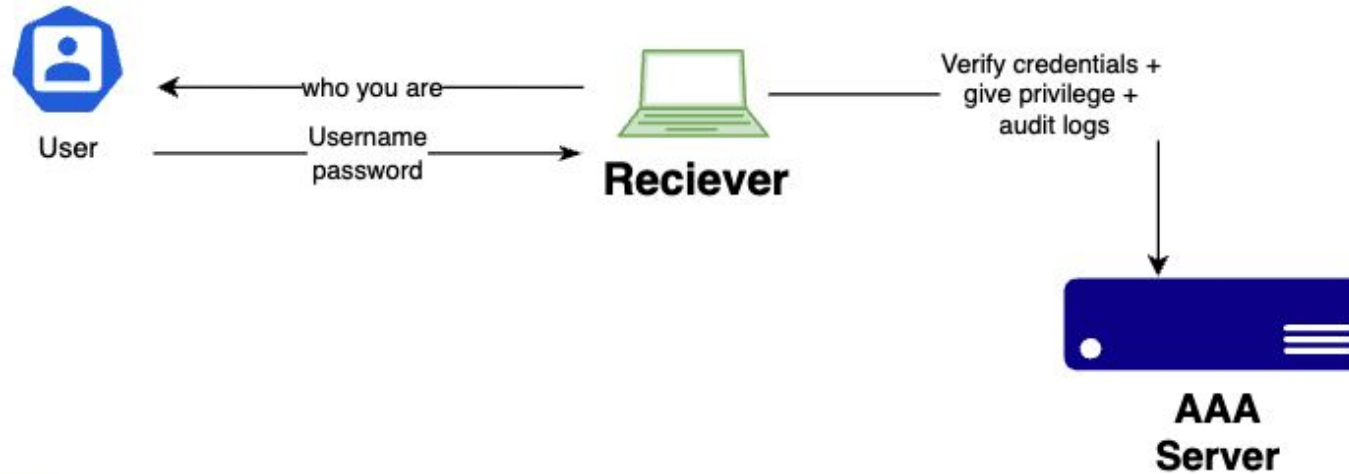
- Data is accurate & not changed



Availability

- Data is accessed when needed, always available & backed up

Authentication • Verify credentials via AAA



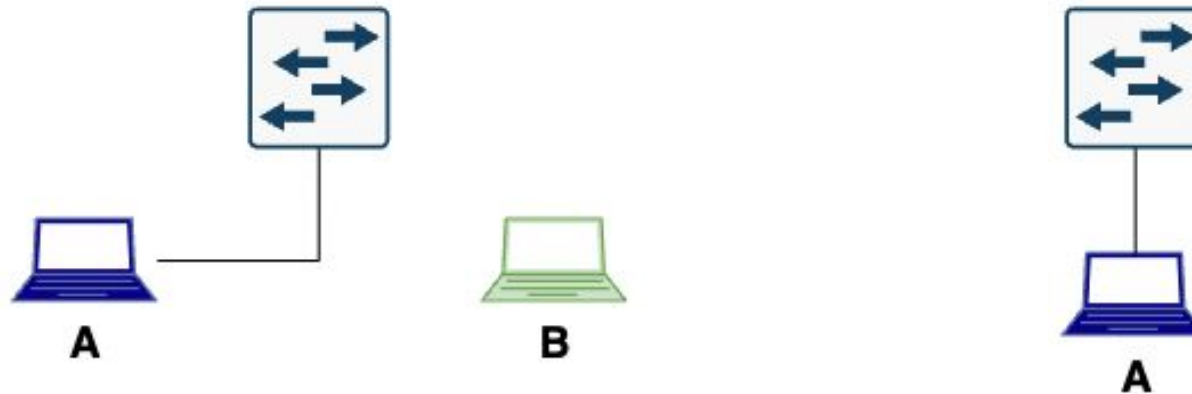
Authorizartion • Determine credential power or privilege of this user

Accounting • Collect information about session, used for auditing and reports

Port security

- Control which mac is learned & can access the port
- If violate then
 - Shutdown
 - Protect (forward from known mac while drop unknown)
 - Restrict (same as protect & send syslog message)

- `switch(config-if)# switchport port-security mac-address sticky`
- `switch(config-if)# switchport port-security mac-address xxxx.xxxx.xxxx`
- `switch(config-if)# switchport port-security violation [protect | restrict | shutdown]`



Standard

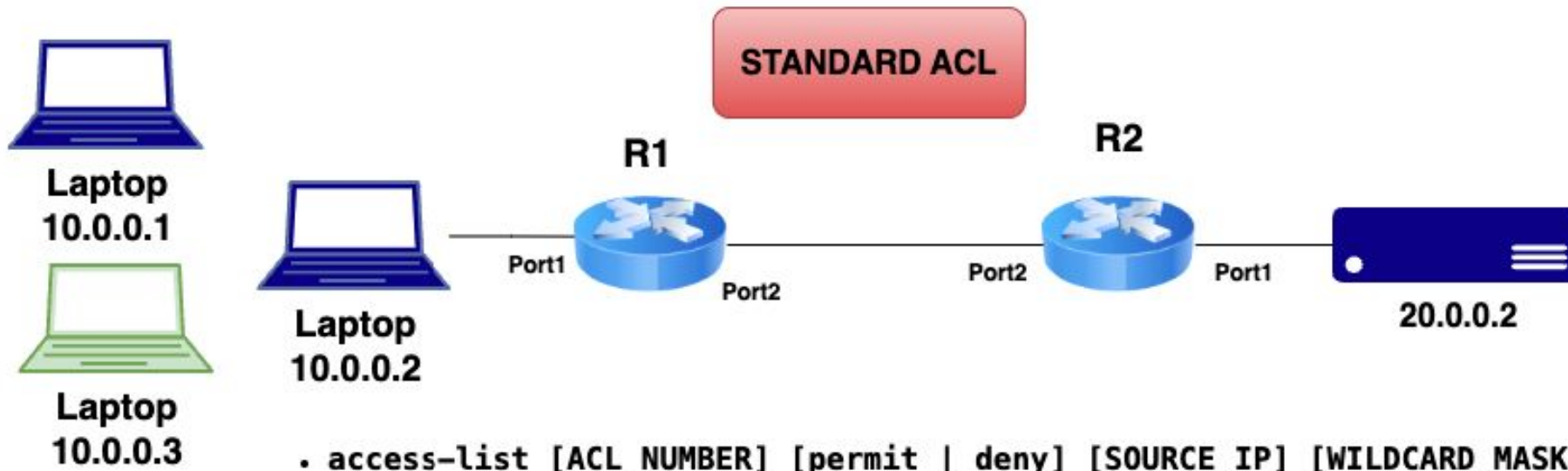
- 1-99 , 1300-1399
- Filter based on source IP
- applied on the router interface inbound, outbound
- Not commonly used
- Applied close as possible to destination

Extended

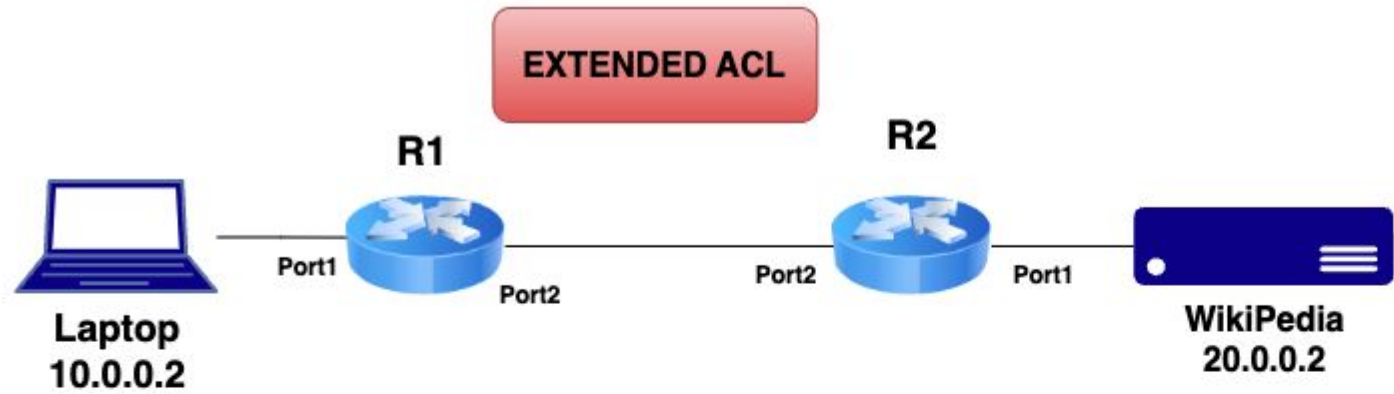
- 100-199 , 200-2699
- Filter based on source IP, destination IP + Source & Destination UDP/TCP
- applied on the router interface inbound, outbound
- More used than standard

WILDCARD MASK

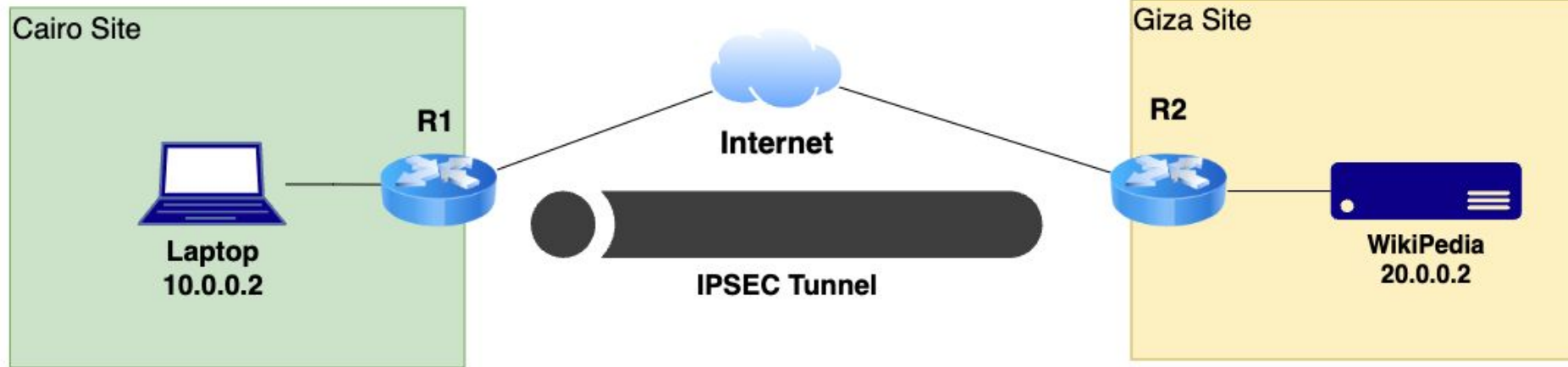
- A wildcard mask is the opposite of a subnet mask. While a subnet mask uses 1 for network bits, a wildcard mask uses 0 for matching bits.
-
- **0 in wildcard mask:** Match this bit exactly (must be the same as in the IP address).
- **1 in wildcard mask:** Ignore this bit (any value is accepted).
-
- 0.0.0.0: Matches a **single IP address** exactly (e.g., 192.168.1.10 0.0.0.0).
- 0.0.0.255: Matches a **subnet range** (e.g., 192.168.1.0/24).
- 0.0.255.255: Matches a **larger range** (e.g., 192.168.0.0/16).



- `access-list [ACL_NUMBER] [permit | deny] [SOURCE_IP] [WILDCARD_MASK]`
- `R2(config)# access-list 1 deny 10.0.0.1 0.0.0.0`
- `R2(config)# access-list 1 deny 10.0.0.2 0.0.0.0`
- `R2(config)# access-list 1 permit any`
- `R2 (config)#interface port 1`
- `R2 (config-if)# ip access-group 1 out`



- `access-list [ACL_NUMBER] or [name] [permit | deny] [PROTOCOL] [SOURCE_IP] [SOURCE_WILDCARD_MASK] [DESTINATION_IP] [DESTINATION_WILDCARD_MASK] [eq PORT_NUMBER]`
- `R1(config)# ip access-list extended BLOCK-WEB`
- `R1(config-ext-nacl)# deny tcp 10.0.0.2 0.0.0.0 20.0.0.2 0.0.0.0 eq 80`
- `R1(config-ext-nacl)# deny tcp 10.0.0.2 0.0.0.0 20.0.0.2 0.0.0.0 eq 443`
- `R1(config-ext-nacl)# permit ip any any`
- `R1 (config)#interface port 1`
- `R1 (config-if)# ip access-group BLOCK-WEB out`



IPSEC Tunnel

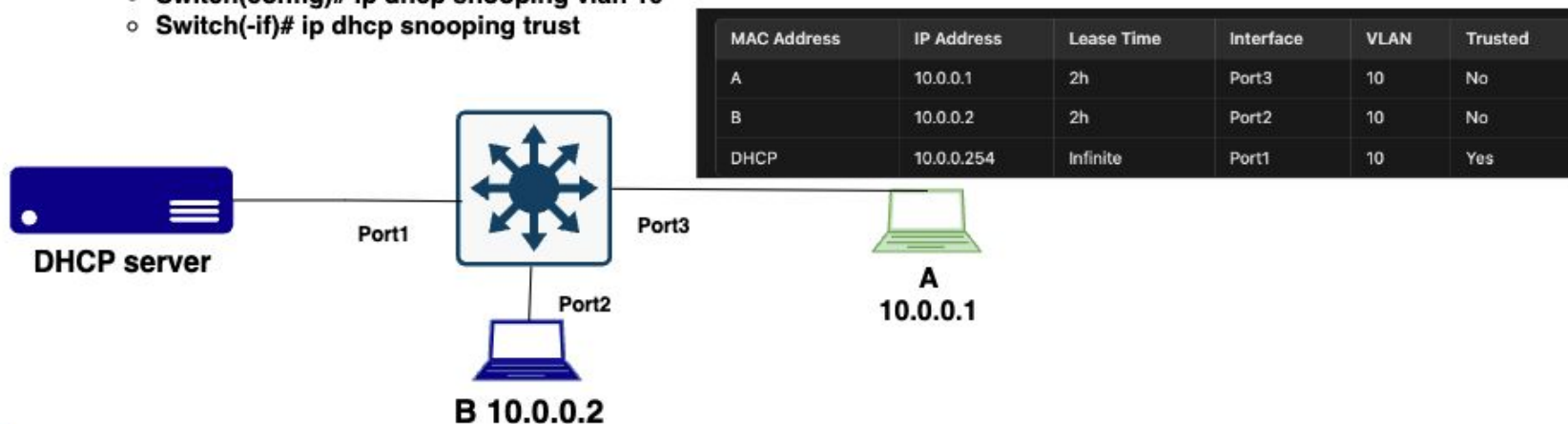
- Designed to secure IP communications by encrypting each IP packet.
- **Phase 1: IKE Phase 1 (ISAKMP Tunnel) - Control**
 - Uses the **Internet Key Exchange (IKE)** protocol to negotiate and set up the initial secure connection
 - Creates a secure channel called the **ISAKMP (Internet Security Association and Key Management Protocol) tunnel**.
- **Phase 2: IKE Phase 2 (IPsec Tunnel) - Data**
 - Uses the **Encapsulating Security Payload (ESP)** protocol for data encryption and integrity.
 - defining the encryption and hashing algorithms.

Phase 1

```
Router(config)# crypto isakmp policy 10
Router(config-isakmp)# encryption aes
Router(config-isakmp)# hash sha
Router(config-isakmp)# authentication pre-share
```

DHCP snooping

- Protects the network from DHCP spoofing attacks by filtering and validating DHCP messages.
- Problem:
 - A Send DHCP Request, Attacker reply by DHCP response that he is the Default GW
- Solution:
 - Configure port 1 as the only trusted port to send DHCP response
 - Switch(config)# ip dhcp snooping
 - Switch(config)# ip dhcp snooping vlan 10
 - Switch(-if)# ip dhcp snooping trust



Dynamic ARP inspection

- Ensures only valid ARP requests and responses are allowed on the network by cross-checking against the DHCP Snooping binding table.
- By default all ports are untrusted
- If violation then error disable
 - Switch(config)# ip arp inspect vlan 10
 - Switch(-if)# ip arp inspection trust | untrust

